

## Carbon fibre sensors revolutionise infrastructure

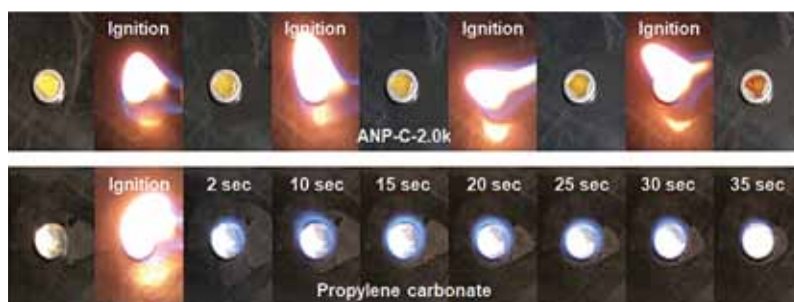


The fibre-reinforced plastic-based triboelectric sensors (FRP-TESs) can detect the vehicle and simultaneously reinforce the surrounding infrastructure, allowing for real-time monitoring and reinforcement of road infrastructure (Credit: Nano Energy 2024, DOI: 10.1016/j.nanoen.2024.109818)

A UNIST research team has revolutionised traffic monitoring with advanced carbon fibre composite sensors. These sensors provide real-time traffic and road condition data while strengthening structures. They facilitate traffic monitoring, accident detection, and prediction by tracking vehicle position, speed, and acceleration. Experiments showed that the FRP, made of carbon fibres, glass fibres, and epoxy, generated electricity when tires passed over it. Even after 10,000 passes, there was no voltage degradation, and its mechanical properties surpassed conventional TES systems. Carbon fibre composites are significantly stronger and lighter than iron, making them ideal for various applications. Future research will integrate FRP's triboelectric mechanism with structural health monitoring (SHM) for intelligent infrastructure.

## Microcrack-free polymer battery extends lifespan

Researchers at the University of Hong Kong have developed new lithium metal batteries with microcrack-free polymer electrolytes. These electrolytes offer extended lifespan, safety, and high electrochemical stability up to 5V, with an ionic conductivity of  $3.1 \times 10^{-5} \text{ S cm}^{-1}$  at high temperatures. The innovation enables batteries to maintain 92.7% capacity retention over 450 cycles at 100°C, outperforming conventional batteries and opening new avenues for high-temperature and fast-charging applications.



The flame tests depict that the new electrolyte membrane (above) is not ignited even with multiple applications of an open flame and resultingly simply turns into coal ash, in contrast with the highly flammable liquid organic solvent (below) (Credit: <https://www.hku.hk/press>)